Formal Epistemology + Philosophy of Science
DAGs: Central Concepts and Causal Structures

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1. Definitions

A **directed acyclical graph (DAG)** consists of a nodes (which stand for variables) and arrows/edges between the nodes, with the additional requirements that the arrows are *unidirectional* and that there is *no cyclical path*.

Variables The basic elements of DAGs, represented graphically by nodes and formally by (italicized) upper case letters. Variables are in our examples typically Boolean (true/false), but they can in principle take all kinds of values.

Arrows Represent causal dependencies between variables

Directed Path A sequence of adjacent arrows in A. The path is **acyclical** if no node occurs twice in it.

Ancestors/Descendants X is an ancestor of Y if there is a directed path from X to Y. In that case, Y is an **descendant** of X. By convention (!), every variable is a descendant of itself.

Parents If X is a parent of Y, there is an arrow from X to Y in the DAG. All parents of Y are (trivially) also ancestors of Y. The parents of Y are typically interpreted as the **direct causes** of Y.

Root and Leaf Node A node without parents is called a **root node**, a node with no descendants apart from itself a **leaf node**.

Endogenous and Exogenous Variables An endogenous variable depends on other variables in the model, an exogenous variable doesn't.

Applying these definitions to Figure 1:

- The parents of C are B and D. A is an ancestor of C, but no parent.
- All nodes in the DAG are descendants of A, including A itself.
- There is exactly one directed path running through all arrows: $A \rightarrow B \rightarrow D \rightarrow C$.
- There is no cyclical path (of course, since it is a DAG!).

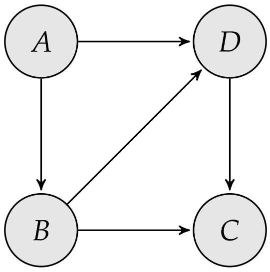


Figure 1: An example of a DAG.

2. Basic Causal Structures

The graphs below the basic DAG structures with three variables: the chain, the common cause ("fork") and the collider.

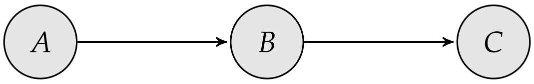


Figure 2: A causal Bayes net in chain form. A and C are independent conditional on B.

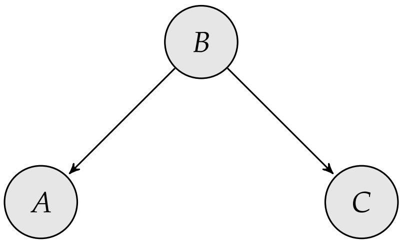


Figure 3: A causal Bayes net with the "fork" or common cause structure. A and C are independent conditional on B.

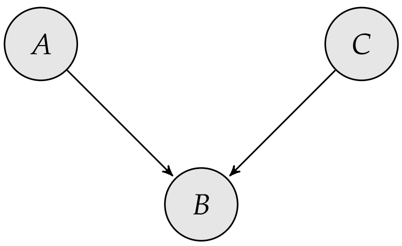


Figure 4: A causal Bayes net with collider structure. A and C are unconditionally independent, but dependent conditional on B.

3. The Causal Markov Condition

DAGs are used in causal inference in combination with a probability distribution over the variables in the DAG. The behavior of the probability distribution is constrained by the "causal knowledge" implemented in the graph as follows:

Causal Markov Condition The probability distribution p is *Markovian* relative to a DAG G if and only if every variable in G is probabilistically independent of all its non-descendants in G, given its parents. In particular, for all $Y \subset \text{Non-Desc}(X)$, $p(X|\text{PA}(X), Y) = p(X|\text{PA}(X))$.

The Causal Markov Condition tells you which (conditional) probabilistic independences are implied by the causal structure of a DAG. A DAG which satisfies the Causal Markov Condition is called a **Causal Bayesian Network (CBN)**, or simply a causal Bayes net.

Notabene: The Causal Markov Condition does *not* license the inference from the absence of a correlation to the absence of a causal connection in the corresponding DAG (this is instead implied by the $\rightarrow$ Faithfulness Condition). Rather, the CMC tells you that the causal structure of a DAG has certain implications on which variables are (conditionally) independent of others.

4. Observation vs. Intervention

The *do*-operator Pearl's *do*-operator is used for describing interventions: the probability distribution of a variable Y given an intervention on X is given by $p(Y|\text{do}(X = x))$. Sometimes also written as $p_{X=x}(Y)$. The probability that Y takes the value y, conditional on the intervention $X = x$, is written as $p(Y = y|\text{do}(X = x))$ or $p_{X=x}(Y = y)$.

Observation vs. Intervention The probability distributions $p(Y|\text{do}(X = x))$ and $p(Y|X = x)$—the distribution of Y conditional on *observing* X—need not agree. A common counterexample occurs when X affects Y not only directly, but also via common causes ("confounders"). The confounding influence is removed if we *intervene* on X, but not if we just observe the value of X (e.g., attendance example).

An example: we are trying to measure the efficacy of a COVID-19 vaccine from general population numbers. Suppose that young people are less likely to get vaccinated than old people. Then, the probability of developing symptoms given vaccination is *confounded* by the effect on age (both on vaccination rates and on the likelihood of COVID-19 symptoms). For this reason, researchers in clinical trial try to obtain unbiased effect size estimates by *intervening* on the putative cause and eliminating the influence of confounders.

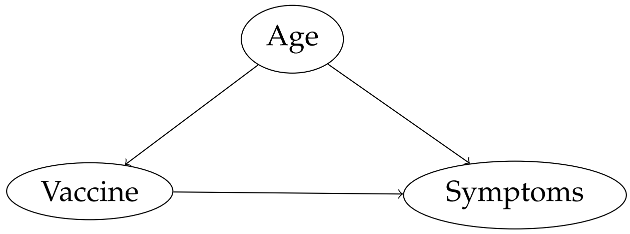


Figure 5: In this example, age is a confounder of the causal effect of vaccination on COVID-19 symptoms.

5. *d*-separation and *d*-connection

The concept of *d*-separation and *d*-connection is a graphical criterion for when two variables can/cannot exert causal influence on each other. It is a ternary relation between sets of variables (e.g., "X is *d*-separated from Y by Z"). For formulating that criterion, we need to understand when an *undirected path* in a DAG (i.e., a path where arrows can not necessarily aligned) is *active*, and when it is *blocked*.[1]

In the simple case, when $Z = \emptyset$, a path is *blocked* if and only if there is a collider on it. (Remember that two variables that are connected by a sequence of edges are unconditionally independent if and only if they are separated by a collider.) It is *active* when there is no collider on it.

This picture is reversed when we consider the presence of a set of variables Z whose values we have observed. Observing a collider creates a dependence between its causes, so if a collider is in Z, the path becomes active. On the other hand, if a non-collider is in Z, this blocks a path (like in a common cause scenario).

Active/Open Path An undirected path between two nodes A and B is *active*, given a set of variables Z if and only if there is no "obstacle" on the path. This means that

- for every collider C on the path, *either* C *or a descendant of* C is in Z; *and*
- *no non-collider* on the path is in Z.

[1] Richard Scheines also gives a somewhat longer tutorial on his website: `https://www.andrew.cmu.edu/user/scheines/tutor/d-sep.html#explanation`.

Blocked Path A undirected path between two nodes A and B is *blocked* by a set of variables Z if and only if it is not active. This can mean any of the following:

- the path contains a collider that is not in Z (and neither is one of his descendants in Z); *or*
- a non-collider on the path is in Z.

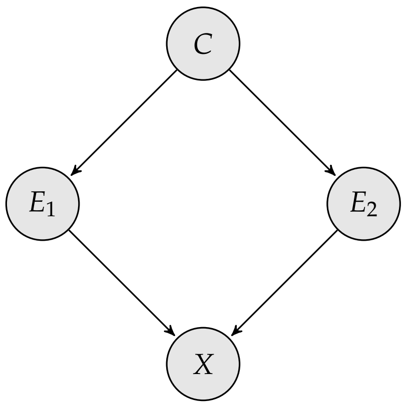
d**-connection** The variables A and B are d-connected by a set of variables Z if there is an active path from A to B (i.e., Z does not block all paths between A and B).

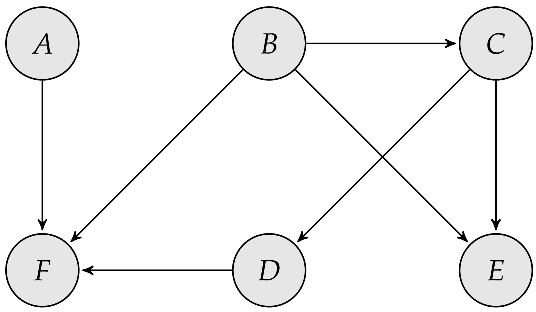
d**-separation** The variables A and B are d-separated by a set of Z if they are not d-connected.

For example, in the above graph, the variable set $\{B, D\}$ d-separates A from C: all paths from A to C are blocked by $\{B, D\}$. This definition can also be generalized to sets of variables (they are d-connected if and only if there is a pair of d-connected variables from the two sets).

The concept of d-separation is important because it is the graphical equivalent to probabilistic independence. In a DAG where the Causal Markov Condition holds, probabilistic independencies can be inferred using the d-separation criterion. Specifically, we can reformulate the Causal Markov Condition as follows: X and Y are conditionally independent on $\mathcal{Z}$ if (not: if and only if) $\mathcal{Z}$ d-separates X and Y.

6. Exercises on (Conditional) Independence in DAGs





Suppose that the above DAGs satisfy the Causal Markov Condition. Then:

(a) Is E_1 independent of E_2, conditional on C?

(b) Is E_1 independent of E_2, conditional on C and X?
(c) Are E and F independent conditional on B?
(d) Are A and C (unconditionally) independent?
(e) Are A and C independent conditional on F?
(f) Are A and C independent conditional on D and F?
(g) Are C and F independent conditional on B and D?

Use the criterion of d-separation/d-connectedness to answer these questions.

7. Internet Resources on DAGs

- Dagitty Browser app for building DAGs: `https://www.dagitty.net/dags.html#`
- Dagitty Manual: `https://www.dagitty.net/manual-3.x.pdf`